

Theories of speech production

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How do the vocal folds vibrate?



Figure: Image of a man sputtering Bronx cheer.

- Try the *Bronx cheer* or *raspberry*.
- VFs are similarly vibrated by the air exhaled out from the lungs.



Neurochronaxic theory of vocal fold vibration

- VFs vibrate in air like strings (mid 18th century).
- Frequency (f_o) of neural impulses generate VF vibration (Husson, 1950).
- This is obsolete.



- Early in the 19th century (Helmholtz and Müller)
- Advanced by van den Berg in the 1950s



- Muscles control the elasticity and tension in the VFs
- Set VFs into vibration and control their (f_o)



- Air pressure from the lungs blows VFs apart.
- Air pressure drops below the VFs when the air escapes through the open glottis.
- VF then close again.



Source-filter theory of vowel production



Figure: Waveform at the glottis changes when it comes out from the lips.

- **Source:** vibration of the VFs
 - Sounds like a buzz, not like speech as we hear it.
- **Filter:** vocal tract between the larynx and the lips.
 - Different configurations due to articulator movement/placement.



Source-filter theory of vowel production



Figure: Vocal tract acts as a filter to change the waveform from the glottis.

- Assumes **independence** between the source of sound and the filter.
- Measure and quantify them separately.



Fundamental frequency (f_o) of VF vibration

- The number of times the vocal folds vibrate per second — they are blown apart and come together.
- Average f_o :
 - Men: ~125 Hz
 - Women: ~100 Hz
 - Children: ~300 Hz or higher (highly variable as they are growing).
- In running speech, average f_o is rarely achieved.



Source function

- VF vibration produces not just one frequency but complex waveform — fundamental frequency and harmonics
 - The filter emphasizes one or the other frequency, which we will see later.
- Drop in amplitude: -12 dB/octave

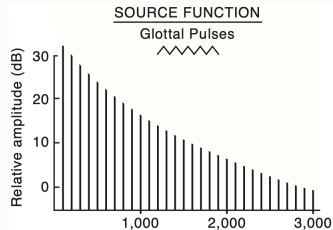


Figure: The spectrum of the glottal source.



Vocal tract: Resonance of a tube open at one end

- Length of an average male vocal tract = 17 cm
- Lowest resonant frequency of the tract (R_1)
$$= \frac{c}{\lambda} = \frac{344\text{ms}^{-1}}{(4*\text{length})} = \frac{34400\text{cm per sec}}{68\text{cm}} \approx 500\text{Hz}$$

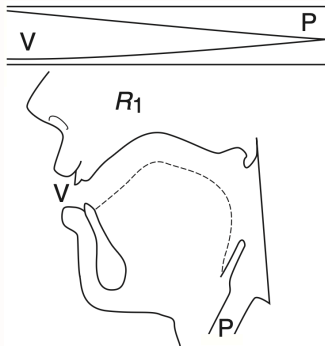


Figure: Vocal tract as a tube open one end; R , resonance; P , point of max pressure; V , point of max velocity.



Resonant frequencies of the vocal tract

- Even multiples: R_2, R_4, R_6 , etc. are not effective.
 - They cancel out at the tube opening.
- Odd multiples: R_1, R_3, R_5 , etc. are additive.
 - 500 Hz, 1500 Hz, 2500 Hz reinforce each-other.

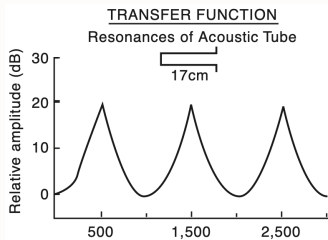


Figure: The frequency response of the vocal tract filter.



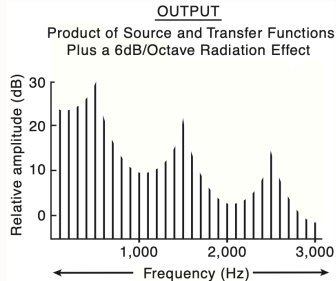


Figure: The spectrum of the speech after source passes through the filter.

Consult Raphael et al. (2011) for further elaboration.



Raphael, L. J., Borden, G. J., & Harris, K. S. (2011). *Speech science primer: Physiology, acoustics, and perception of speech* (6th ed.). Lippincott Williams & Wilkins.

